

Haryana Government Digital Repository

Project Overview & Technical Brief

Government of Haryana

1. Purpose

The Haryana Government Digital Repository is a bilingual (Hindi / English), role-based digital document management portal built for the Government of Haryana. It enables government departments to upload, review, approve, and publish legal documents such as Acts, Rules, Notifications, Circulars, Orders, and Policies in a secure, auditable, and searchable environment. Citizens can access published documents through a public portal without logging in.

2. System Architecture

The system is split into two independent applications that communicate over a REST API.

Layer	Technology
Frontend	React 19 + Vite
Backend API	FastAPI (Python 3.12)
Database	MySQL
ORM	SQLAlchemy 2.0
AI / Semantic Search	Ollama LLM + ChromaDB Vector Store
File Storage	Local filesystem (uploads/)

3. Key Features

Document Management

- Upload PDF and DOCX documents with rich metadata (title, department, type, year, tags).
- Automatic text extraction via PyMuPDF; scanned files fall back to Tesseract OCR (English + Hindi Devanagari).
- Version management with multiple versions tracked per Act.
- Department-level cross-referencing between related documents.

Approval Workflow

- Uploaders submit documents; Approvers review and approve or reject with written remarks.
- Act Parts (sections, schedules, annexures, appendices, forms) have their own approval lifecycle.
- Full audit trail of every approval decision.

Semantic Search & AI

- Every uploaded page is embedded and indexed in ChromaDB (vector database).
- Natural-language queries answered via a RAG pipeline using Ollama LLM with source references.
- Documents are automatically summarised by the llama3.2 model on upload.

Security & Authentication

- Username + password with CAPTCHA for all staff roles.
- Super Admin: mobile OTP-only login (no password).
- RSA-encrypted login payloads; bcrypt password hashing; JWT Bearer tokens (30-min expiry).
- 6-month password-expiry policy; first-login mobile OTP verification.
- Per-department per-role user caps to prevent role overcrowding.

Bilingual UI

- Full Hindi and English support via i18next.
- In-browser Hindi keyboard component for Devanagari text entry.
- Tesseract OCR configured for both English and Hindi language packs.

Notifications

- BSNL SMS API (DLT-compliant) for OTP delivery.
- SMTP email for password-reset OTPs.
- In-app browser notifications for document status changes.

Audit & Compliance

- Every significant action logged with actor, IP address, timestamp, and field-level diff.
- Dedicated Auditor role for read-only audit log access and CSV export.

Analytics

- Charts (Recharts + D3.js): document counts, department activity, approval rates, trends.
- Role-specific analytics views for CSO and Admin dashboards.

4. User Roles

Role	Access Level	Key Responsibilities
Citizen	Public, read-only	Search and view published legal documents
Uploader	Authenticated	Upload documents, manage drafts, submit for approval
Approver	Authenticated	Review, approve or reject documents and act parts
Nodal Officer	Authenticated	Manage users and documents within assigned departments
Admin	Authenticated	User management, department config, role assignment
Super Admin	OTP-authenticated	Full system control across all departments and users
CSO	Authenticated	System-wide analytics and oversight
Auditor	Authenticated	Read-only access to full audit logs and CSV export

5. Document Types Supported

- Act
- Amendment
- Notification
- Circular
- Policy
- Rules & Regulations
- Order / Gazette
- Bye Laws
- Miscellaneous

6. Act Parts Module

Legal Acts can be broken down into structured parts, each with its own independent submission and approval lifecycle. Each part type supports file uploads (PDF / DOCX).

Part Type	Description
Sections	Core legal text with optional chapter grouping
Schedules	Supplementary schedules attached to the Act
Annexures	Detailed annexure documents
Appendices	Supporting appendix material
Forms	Official forms prescribed under the Act

7. Technology Stack

Frontend Libraries

Library	Purpose
React 19	UI framework
Vite	Build tool
React Router 7	Client-side routing
Axios	HTTP client
PDF.js	In-browser PDF rendering
Mammoth	DOCX to HTML conversion
ExcelJS	Excel data export
Recharts + D3.js	Analytics charts
JSEncrypt	RSA payload encryption
i18next	Hindi / English internationalisation

Backend Libraries

Library	Purpose
FastAPI	REST API framework
SQLAlchemy 2	ORM with MySQL via PyMySQL
PyMuPDF	PDF text extraction
Tesseract OCR	Scanned document OCR (English + Hindi)
python-docx	DOCX processing
Ollama	Local LLM: llama3.2 + nomic-embed-text
ChromaDB	Vector store for semantic search
bcrypt + PyJWT	Password hashing and JWT auth
cryptography	RSA key management
BSNL SMS	SMS OTP delivery (DLT-compliant)

8. Data Flows

Document Upload to Publication

- **Step 1:** Uploader selects PDF or DOCX and fills in metadata (title, department, type, year, tags).
- **Step 2:** Backend extracts text using PyMuPDF; falls back to Tesseract OCR for scanned files.
- **Step 3:** Ollama LLM (llama3.2) generates an automatic summary of the document.
- **Step 4:** Document pages are embedded and stored in ChromaDB for semantic search.
- **Step 5:** Approver reviews in the portal and approves or rejects with remarks.
- **Step 6:** On approval, the document is published and visible to citizens.

Semantic Search (RAG Pipeline)

- **Step 1:** User types a natural-language query in the search bar.
- **Step 2:** ChromaDB performs cosine-similarity search across all indexed document pages.
- **Step 3:** The most relevant pages are passed as context to the Ollama LLM.
- **Step 4:** LLM generates a precise answer with source document references.

9. Security Highlights

Security Control	Detail
RSA-encrypted login	Credentials encrypted in-browser before transmission
CAPTCHA	Blocks automated login attacks
bcrypt password hashing	Passwords never stored in plaintext
JWT tokens	30-minute expiry; Bearer token in all API requests
OTP (10-min expiry)	Mobile OTP for admin login and password reset
Password expiry policy	6-month forced password rotation for all staff
Role-based access control	Enforced at the API layer via FastAPI dependencies
Per-dept role caps	Prevents overcrowding of roles within a department
Full audit trail	Actor, IP address, timestamp, and field-level diff logged
Path-traversal protection	File-serve endpoints reject paths containing / or ..